

Day 17 — Provider Evaluation Report

Email classification: Gemini vs Groq

Workflow Overview

This n8n workflow runs an identical 10-email classification test dataset through two LLM providers in parallel — Google Gemini (gemini-3.6-flash) and Groq (openai/gpt-oss-120b) — using the same classification prompt. Each output is scored against an expected label, the two branches are merged, and a summary report is generated.

Evolution: What Changed

In-code comments on the scoring nodes (Latency for Gemini, Latency2 for Groq) show this is a revised version of an earlier, buggier evaluation:

- **Cost estimation** — replaced a flat/placeholder cost constant with a character-count-based token estimate (~4 chars/token), falling back to real tokenUsage/usage_metadata from the model response when available.
- **Error handling** — failed calls are now explicitly detected and marked ERROR with the captured error message, instead of being silently scored as a FAIL with an empty output.
- **Pass/fail logic** unchanged: a case passes if there is no error, an expected label exists, and the model output contains that label (case-insensitive substring match).

Current Run Results

Provider	Accuracy	Errors	Avg Latency	Est. Cost
Gemini	80.0%	0	0 ms	\$0.00002
Groq	90.0%	0	0 ms	\$0.00001

Recommendation: **Groq** — it outperformed Gemini on both accuracy (90.0% vs 80.0%) and estimated cost (\$0.00001 vs \$0.00002) in this run.

Flagged Issue: Recommendation Logic

The Report node's own recommendation logic only compares raw pass counts (gPassed >= qPassed ? "Google Gemini" : "Groq"), defaulting to Gemini on a tie and ignoring cost and latency entirely, even though both are computed. In this run Groq had more passes (9 vs 8), so the output happened to be correct, but the logic is fragile and does not reflect a genuinely weighted evaluation across accuracy, cost, and latency.

Suggested Next Step

Rework the recommendation logic to weigh accuracy, cost, and latency together (e.g. a scoring formula or explicit tie-breaking rule) rather than relying on a single pass-count comparison.